SPORTS ANALYTICS QUANTITATIVE RESEARCH NOTE

# ELO & POWER RATING MODELS

## Forecasting Moneyline, Point Spreads and Points Over/Under

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NFL NBA COLLEGE FOOTBALL

A mathematical derivation, league-specific scoring design, probability engine, production architecture and testing protocol.

October 19, 2026

Research design and illustrative simulations; not live betting recommendations.

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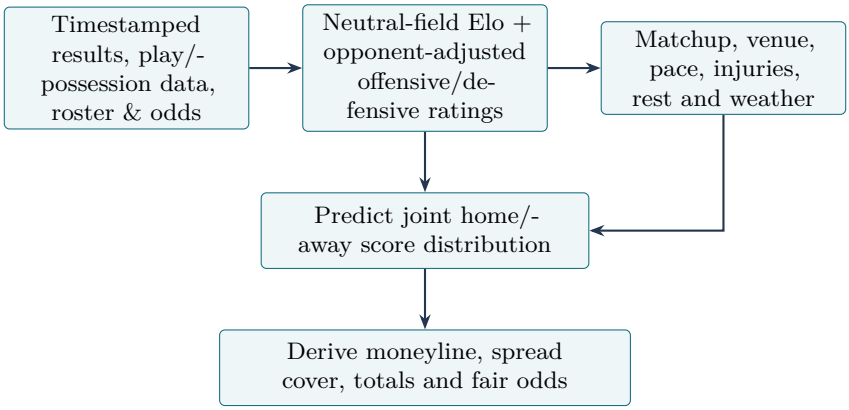
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# 1 Executive model specification

The objective is to estimate a *complete pregame distribution* of home and away final scores rather than just a winner. From one predictive distribution, derive moneyline probability, all candidate handicaps, all total-point thresholds, fair prices and uncertainty bands. Maintain separate fitted parameter sets for the NFL, NBA and college football (CFB).

**Foundational distinction.** Plain Elo is a relative team-strength and win-probability model. Two games can have the same Elo differential but radically different totals (e.g., 17–14 versus 38–35). An O/U forecast requires a separate offensive/defensive scoring model and a pace/possession model. A production design combines Elo, adjusted points-based power ratings and context into a jointly calibrated score distribution.




## 1.1 Definitions and orientation

Set $H$ = home team, $A$ = away team. Let $Y_H, Y_A$ denote their final scores under the market's settlement convention; then

$$M = Y_H - Y_A, \qquad T = Y_H + Y_A, \qquad Y_H = \tfrac{T+M}{2}, \quad Y_A = \tfrac{T-M}{2}. \tag{1}$$

Positive $M$ means a home victory; $s$ is an offered *home-team spread line*, so home cover is $M + s > 0$. A total line $L$ is over when $T > L$. When lines are integers, model the push event rather than treating it as a loss. All target probabilities must correspond to the same clock, overtime and grading rules as the collected sportsbook market.

| Model output | Mathematical target | Primary model ingredient | Consumer |
|---|---|---|---|
| Home moneyline | $\Pr(H \text{ wins})$ | Full margin/final-score law | Fair moneyline price |
| Home spread $s$ | $\Pr(M + s > 0)$ | Margin distribution | Cover odds; alt spreads |
| Over $L$ | $\Pr(T > L)$ | Total distribution | Over/under; alt totals |
| Correct team scores | $\Pr(Y_H = h, Y_A = a)$ | Joint score distribution | Correlation, scenario analysis |

# 2 Elo: strength estimation from sequential games

## 2.1 The classic Elo probability equation

Each team $i$ has a neutral-location rating $R_{i,t}$ before game $t$. At the prediction step define

$$D_t = R_{H,t} - R_{A,t} + H_{\ell,t} + x_t^\top \gamma_\ell, \qquad p_{E,t} = \frac{1}{1 + 10^{-D_t/c_\ell}}. \tag{2}$$

$H_{\ell,t}$ is home advantage in *Elo units*, $x_t$ contains known rest, player, travel and other pregame changes, $\ell$ indexes the league, and the rating scale $c_\ell = 400$ is a conventional starting point rather than a universal law. Estimate or

calibrate the scale on historical held-out games. All $R_i$ remain *neutral*; never permanently add home advantage to a team's persistent rating.

An equivalent log-odds relationship is

$$\log \frac{p_E}{1-p_E} = \frac{\ln 10}{c_\ell} D. \tag{3}$$

A 100-point Elo gap maps to $1/(1+10^{-0.25}) \approx 64.0\%$ at $c = 400$. By construction, this is a probability of *the Elo model's defined game result*; a joint score model still needs calibration for ties and overtime.

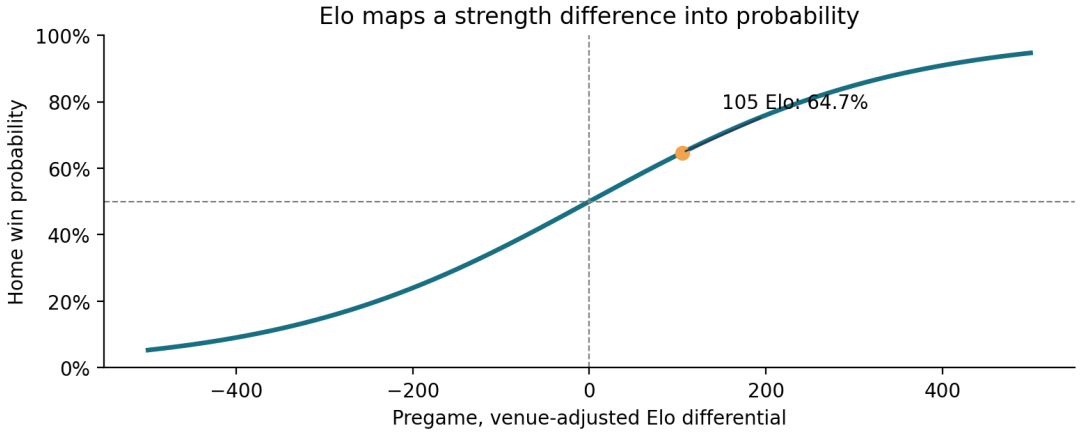



**Figure 1:** Standard Elo logistic transformation; illustrative, not an empirical fit.

## 2.2 Postgame updating and surprise

For a conventional zero-sum two-team update, let $S_H = 1$ for a home win, 0.5 for an Elo-defined tie, and 0 for a home loss. With $S_A = 1 - S_H$,

$$\Delta R_H = K_\ell \, g_\ell(|M_t|, D_t, S_H)\,(S_H - p_{E,t}), \qquad R_{H,t+1} = R_{H,t} + \Delta R_H, \quad R_{A,t+1} = R_{A,t} - \Delta R_H. \tag{4}$$

$K_\ell$ controls adaptation speed; $g_\ell$ is an optional margin-of-victory (MOV) modifier. Start with $g \equiv 1$; compare with a regularized MOV specification. One candidate is $g \propto \log(1 + |M|/m_0)$ multiplied by a favorite/underdog adjustment estimated on past games. Simply rewarding all blowout wins equally can introduce a rating drift because favorites tend to win by larger margins [1, 2].

**Time-order discipline:** compute $p_{E,t}$ from the rating *before* game $t$, store the frozen forecast, and only then apply Equation (4). Simultaneous games in a week or slate should be forecast from the appropriately frozen pregame information set.

## 2.3 Season transition and changing rosters

A basic offseason prior is

$$R_{i,\text{new}} = 1500 + \rho_\ell (R_{i,\text{old}} - 1500) + z_i^\top \delta_\ell, \qquad 0 \le \rho_\ell \le 1. \tag{5}$$

$\rho_\ell$ discounts old results; $z_i$ captures genuinely forward-looking roster, coaching and personnel information known by that date. Do not use end-of-season player summaries for predictions made earlier in the season. CFB needs materially stronger program/roster priors than a bare return-to-mean update because membership, schedule strength and annual roster turnover differ from pro leagues.

# 3 Points-based power ratings: the bridge to spreads

## 3.1 Opponent-adjusted margin model (Massey/ridge)

A single-dimension rating $r_i$ measures projected points better than average on a neutral field. For historical game $g$ let $m_g = y_{H,g} - y_{A,g}$. Model

$$m_g = r_{H(g)} - r_{A(g)} + h_\ell + x_g^\top \beta_\ell + \epsilon_g. \tag{6}$$

After subtracting venue and contextual components, define response $\widetilde{m}_g$. Construct an incidence matrix $X$ with $X_{g,H} = +1$, $X_{g,A} = -1$, zero otherwise, and game weights $w_g$ (recency, relevance and data quality). Then solve

$$\widehat{\boldsymbol{r}} = \arg\min_{\boldsymbol{r}:\, \mathbf{1}^\top \boldsymbol{r}=0} (\widetilde{\boldsymbol{m}} - X\boldsymbol{r})^\top W (\widetilde{\boldsymbol{m}} - X\boldsymbol{r}) + \lambda \|\boldsymbol{r} - \boldsymbol{r}_0\|_2^2. \tag{7}$$

This is an opponent-adjusted, ridge-regularized form of points-based ranking, related to Massey's least-squares framework [4]. The constraint fixes the arbitrary overall level; $\lambda > 0$ and pregame prior $\boldsymbol{r}_0$ stabilize lightly connected college-football schedules. For $\lambda = 0$ and connected schedules, the normal-equation coefficient matrix is $X^\top W X$, the weighted graph Laplacian.

Without the sum-to-zero constraint, a regularized closed-form solution (for a centered zero prior or suitably specified identifiable parameterization) is

$$\widehat{\boldsymbol{r}} = (X^\top W X + \lambda I)^{-1} (X^\top W \widetilde{\boldsymbol{m}} + \lambda \boldsymbol{r}_0). \tag{8}$$

In practice use QR, Cholesky or a stable numerical least-squares solver, rather than explicitly forming an inverse; impose identifiability consistently if estimating an intercept.

**Interpretation:** if $r_H = +4.0$, $r_A = -1.0$, and estimated home edge $h = 1.6$ points, then $\widehat{M} = 4 - (-1) + 1.6 = 6.6$ points before further matchup effects. The unit of $r$ is points, unlike Elo's abstract rating units.

## 3.2 Dynamic ratings rather than static season rankings

A weighted retrospective fit is the simplest implementation. A more principled online alternative is a state-space model:

$$r_{i,t} = r_{i,t-1} + \eta_{i,t}, \qquad \eta_{i,t} \sim \mathcal{N}(0, q_{\ell,i,t}), \tag{9}$$

$$m_t = r_{H,t} - r_{A,t} + h_\ell + x_t^\top \beta + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, v_\ell). \tag{10}$$

A Kalman filter (or Bayesian dynamic hierarchical model) propagates both *estimates and uncertainty*. Large $q$ after QB/roster shocks lets ratings adapt quickly; low $q$ on stable weeks prevents overreaction. An NFL weekly refresh, NBA daily refresh and CFB weekly refresh can use the same general framework with different hyperparameters.

## 3.3 Deriving team offenses and defenses (essential for O/U)

Margin-only power ratings identify *difference*, not scoring level. Let $o_i$ represent offense above league baseline, $d_i$ represent defensive quality (a larger $d_i$ *suppresses* opponent scoring), $\mu_\ell$ baseline points per team, and $h_H, h_A$ home/away effects. Fit both score equations jointly:

$$y_{H,g} = \mu_{\ell,g} + o_{H,g} - d_{A,g} + h_H + x_{H,g}^\top \theta + \epsilon_{H,g}, \tag{11}$$

$$y_{A,g} = \mu_{\ell,g} + o_{A,g} - d_{H,g} + h_A + x_{A,g}^\top \theta + \epsilon_{A,g}. \tag{12}$$

Enforce $\sum_i o_i = \sum_i d_i = 0$ within the relevant population; apply ridge/hierarchical priors. Use a link function or possession-based scale suited to the league (Sections 4–6) if linear score predictions can become negative. A single rating equivalent to $o_i + d_i$ measures strength but cannot uniquely reconstruct offense and defense, so fit the latter from actual score/efficiency observations.

The game forecasts are $\widehat{\mu}_H, \widehat{\mu}_A$ and satisfy the accounting identities

$$\widehat{\mu}_M = \widehat{\mu}_H - \widehat{\mu}_A, \qquad \widehat{\mu}_T = \widehat{\mu}_H + \widehat{\mu}_A. \tag{13}$$

Do *not* independently manufacture an unrelated total and two unrelated team-score forecasts.

# 4 NFL model: drive frequency and scoring efficiency

## 4.1 Why yards or raw points per game are insufficient

Football's score is a consequence of drive count, drive starting field position, offensive strength, defensive strength, quarterback availability, turnovers and non-offensive scoring. Fit an Elo prior and a dynamic points rating but use *drive-level* scoring as the preferred totals engine. Public nflverse play-by-play, schedules and supporting data can underpin historical estimation [6, 7].

Let $N_H, N_A$ be future offensive drive counts (not necessarily independent). Estimate conditional drive count from matchup pace, expected plays, pass rate, possession length, clock rules and game-state interactions. For drive $j$ of team $i$, use a mutually exclusive outcome $Z_{ij}$ from an explicitly defined scoring alphabet (e.g., 0, 3, 6, 7, 8 points as appropriate after the full conversion outcome). One model is

$$\Pr(Z_{ij} = k \mid \mathcal{I}_t) = \pi_{i,k}(f_i, \text{opposing defense}, \text{field position}, \text{weather}, \text{QB}, \text{game state}), \quad \sum_{k \in \mathcal{K}} \pi_{i,k} = 1. \tag{14}$$

Handle safety, defensive/special-teams scores, penalties and overtime separately or with richer event states; the toy alphabet is not an exhaustive enumeration of football rules. Then

$$Y_i = \sum_{j=1}^{N_i} Z_{ij} + D_i + ST_i + \text{other scored events}, \qquad \mathbb{E}[Y_i] \approx \mathbb{E}[N_i]\mathbb{E}[Z_i] + \mathbb{E}[D_i + ST_i] \tag{15}$$

under a simplified conditional-homogeneity/independence approximation; otherwise evaluate expectations under the full simulator. Opponent-adjusted per-drive production can be modeled with an exponential or multinomial link so inferred scoring rates are nonnegative.

## 4.2 Football-specific covariates

| Feature family | Margin impact | Total-points impact |
|---|---|---|
| Starting QB and protection | Team-specific offensive edge | Touchdown/turnover/drive-success mix |
| Off/def drive efficiency | Score differential | Both teams' scoring distributions |
| Play pace, drives, pass rate | Usually modest direct effect | Primary volume lever |
| Weather, roof, wind | Passing/kicking differential | Scoring distribution, variance |
| Rest, travel, short week | Temporary team-quality effect | Potential pace and efficiency effect |
| Injuries / defensive matchup | Matchup-specific edge | Scores, especially red-zone efficiency |
| Non-offensive TD risk | Upset/tail risk | Extra discrete scoring tail |

Estimate interactions rather than applying the same "wind points adjustment" to every offense: severe wind changes an outdoor pass-heavy matchup differently from an indoor run-heavy one. Keep all features as-of kickoff or the forecast timestamp.

# 5 NBA model: possessions times efficiency

## 5.1 Estimate shared game pace

NBA points per game confound pace and efficiency. The NBA defines offensive rating as points scored per 100 possessions and pace as possessions per 48 minutes [5]. Denote $P$ = estimated possessions per team in regulation. A simple baseline is

$$\widehat{P} = \omega \frac{P_H + P_A}{2} + (1 - \omega) P_\ell + z_t^\top \phi, \tag{16}$$

where $\omega$ is out-of-sample fitted, $P_\ell$ is the contemporary league pace and $z$ covers lineup, rest, fatigue and style. A predictive distribution for $P$ matters: one realized pace shock raises or lowers *both* teams' totals, producing positive score covariance. For independent identically distributed per-possession point yields $X_{ij}$ that are independent of $P$, the random-sum identity gives

$$\operatorname{Var}\left(\sum_{j=1}^{P} X_{ij}\right) = \mathbb{E}[P]\operatorname{Var}(X_{ij}) + \operatorname{Var}(P)\big(\mathbb{E}[X_{ij}]\big)^2. \tag{17}$$

This separates possession-level randomness from the extra uncertainty due to the total number of possessions. The identical logic applies to a simplified random-count football-drive model, with drives replacing possessions.

### 5.2 Match offense to opponent defense

Express baseline league points per possession by $q_\ell > 0$. A useful positive-rate model is

$$q_H = q_\ell \exp(o_H - d_A + u_H), \tag{18}$$

$$q_A = q_\ell \exp(o_A - d_H + u_A), \mathbb{E}[Y_H^{\text{reg}}] \qquad \approx \mathbb{E}[Pq_H], \qquad \mathbb{E}[Y_A^{\text{reg}}] \approx \mathbb{E}[Pq_A]. \tag{19}$$

Here $o, d, u$ are log-scale ratings and game-specific changes. $d_i > 0$ signals better defense. If $P$ and $q_i$ are conditionally independent, simplify to $\mathbb{E}[P]\mathbb{E}[q_i]$; otherwise keep the covariance term. A practical fitter uses possession-weighted scoring observations with a log-link and temporal shrinkage. For example, with illustrative $P = 100$, $q_\ell = 1.15$, $o_H = .040$, $d_A = -.015$, $o_A = -.010$, $d_H = .025$, and zero extra context, the home and away means are approximately $100(1.15)e^{.055} = 121.50$ and $100(1.15)e^{-.035} = 111.04$ points, respectively. Consequently $\mu_M \approx 10.46$ and $\mu_T \approx 232.54$ before overtime. These are hypothetical model inputs, not current NBA estimates.

**Roster replacement:** roster-availability changes should explicitly adjust $o_i, d_i$ using projected on-court minutes and estimated player contributions; old win/loss Elo alone lags major trades, load management and injuries. Prior NBA models have blended Elo with roster/player models and rest/travel adjustments [3].

### 5.3 Regulation, overtime and calibration

Sportsbook NBA full-game markets commonly include overtime, but verify the individual market. Simulate regulation first; when tied, simulate overtime possession/scoring conditional on the teams' lineup, fatigue and shortened period. Alternatively calibrate directly to final-score historical data with an overtime indicator/generator. Do not use a regulation-only expectation for an overtime-included totals market.

## 6 College football: hierarchical strength of schedule

CFB's large and incompletely connected opponent network produces weak early-season pure Elo rankings. Use division/conference-strength priors and continuously opponent-adjusted scoring rates. CFBD exposes game and advanced historical data suitable for constructing features [8].

### 6.1 Hierarchical preseason prior

One specification is

$$r_{i,0} \mid c(i) \sim \mathcal{N}(\mu_{c(i)} + z_i^\top \delta, \tau^2_{c(i)}), \tag{20}$$

$$\mu_c \sim \mathcal{N}(\mu_{\text{FBS}}, \tau^2_{\text{conf}}), \tag{21}$$

$$r_{i,t} = r_{i,t-1} + \eta_{i,t}. \tag{22}$$

$c(i)$ is conference/division grouping; $z_i$ can contain last year's opponent-adjusted strength, returning production, QB continuity, transfer/recruiting indicators and coaching change as known *at prediction time*. A substantial prior variance flags genuine uncertainty in teams with limited interconference evidence. Use a distinct calibration or interaction for FBS–FCS games and validate their data coverage.

### 6.2 Efficiency and drive volume

Let $\widehat{N}$ be a game's expected offensive possessions/drives per team, acknowledging that the realized counts depend on one another. Model each team's expected points per drive $e_i$ via opponent-adjusted offense, defense, field position, explosive plays and finishing-drives indicators. Then

$$\widehat{\mu}_i \approx \widehat{N}_i \, \widehat{e}_i + \widehat{\text{non-offensive points}}_i. \tag{23}$$

CFB totals require more than an opponent-adjusted margin: a methodical low-tempo 28–14 matchup and a fast 49–35 matchup may have comparable favorite strength but drastically different totals. For illustration, an expected 12.5 drives per team with 2.7 and 1.9 expected points per drive produces means of 33.75 and 23.75, margin +10.0 and total 57.5 points, ignoring non-offensive scoring. This example exhibits why drive count is an independent O/U input. For post-regulation grading, implement the applicable overtime rules for each season instead of using the NFL overtime model.

**CFB leakage trap:** end-of-season SRS, cumulative ratings, strength-of-schedule and returning-production data can encode games that had not yet been played. Reconstruct every feature using only the information published before each scheduled kickoff.

## 7 From projected scores to probabilities and fair prices

### 7.1 Quick baseline: bivariate normal approximation

As a first-stage benchmark (most useful for NBA and as an approximate football baseline), suppose

$$\begin{bmatrix} Y_H \\ Y_A \end{bmatrix} \sim \mathcal{N}_2\left( \begin{bmatrix} \mu_H \\ \mu_A \end{bmatrix}, \begin{bmatrix} \sigma_H^2 & \rho\sigma_H\sigma_A \\ \rho\sigma_H\sigma_A & \sigma_A^2 \end{bmatrix} \right). \tag{24}$$

Then $M, T$ are jointly normal with

$$\mu_M = \mu_H - \mu_A, \qquad \sigma_M^2 = \sigma_H^2 + \sigma_A^2 - 2\rho\sigma_H\sigma_A, \tag{25}$$

$$\mu_T = \mu_H + \mu_A, \qquad \sigma_T^2 = \sigma_H^2 + \sigma_A^2 + 2\rho\sigma_H\sigma_A, \tag{26}$$

$$\operatorname{Cov}(M, T) = \sigma_H^2 - \sigma_A^2. \tag{27}$$

The last identity is worth noting: positive team-score correlation inflates total variance and decreases margin variance, and margin/total dependence need not vanish when team-score variances differ.

Under the continuous approximation, with $\Phi$ the standard normal CDF,

$$\Pr(\text{home win}) \approx \Phi(\mu_M/\sigma_M), \tag{28}$$

$$\Pr(\text{home covers } s) \approx \Phi((\mu_M + s)/\sigma_M), \tag{29}$$

$$\Pr(\text{over } L) \approx \Phi((\mu_T - L)/\sigma_T). \tag{30}$$

For integer-valued scores and integer line handling, compute probabilities from discrete simulated totals/margins. A normal continuity correction for $\Pr(T \geq 48)$ is $\Pr(T_{\text{continuous}} > 47.5)$; do not use it indiscriminately when a push can occur.

### 7.2 Preferred production engine: a coherent joint score simulator

Draw $B$ plausible futures conditional on pregame information $\mathcal{I}_t$:

$$(\theta^{(b)}, \text{pace}^{(b)}, \text{state}^{(b)}) \sim p(\theta, \text{pace}, \text{state} \mid \mathcal{I}_t), \qquad (Y_H^{(b)}, Y_A^{(b)}) \sim p_\ell(\cdot \mid \theta^{(b)}, \text{pace}^{(b)}, \text{state}^{(b)}). \tag{31}$$

Derive the event probabilities using the *same* score draws:

$$\widehat{p}_{H,\mathrm{reg}} = B^{-1}\sum_b \mathbf{1}[M^{(b)} > 0], \tag{32}$$

$$\widehat{p}_{\mathrm{cover}}(s) = B^{-1}\sum_b \mathbf{1}[M^{(b)} + s > 0], \tag{33}$$

$$\widehat{p}_{\mathrm{over}}(L) = B^{-1}\sum_b \mathbf{1}[T^{(b)} > L], \tag{34}$$

$$\widehat{p}_{\mathrm{push}}(L) = B^{-1}\sum_b \mathbf{1}[T^{(b)} = L]. \tag{35}$$

For a full-game moneyline, resolve tied regulation scores with the league- and season-specific overtime simulator; replace $\widehat{p}_{H,\mathrm{reg}}$ with the probability of the *graded final result*. Treat NFL possible final ties in accordance with the market's settlement terms. Standard error of a simulated Bernoulli probability is approximately $\sqrt{\widehat{p}(1-\widehat{p})/B}$ conditional on a fixed fitted model; this excludes parameter/model uncertainty.

### 7.3 Choose the correct score family

| League | Baseline | Preferred refinement |
|---|---|---|
| NFL | Approximate bivariate normal for initial validation | Shared drive count, categorical drive scores, non-offensive TDs, state-dependent drives, OT |
| NBA | Correlated normal scores or possession count times efficiency | Hierarchical count/possession process, shared pace, player lineups, OT |
| CFB | Bivariate normal with broad, conditional variance | Hierarchical team strength, conference schedule graph, discrete drives, CFB OT |

A continuous normal can assign impossible negative scores and misses discrete football scoring clusters. Use it as a transparent benchmark, not the definitive high-resolution alt-totals engine.

### 7.4 Mapping fair probabilities to bookmaker prices

For no-push two-outcome markets with model probability $p$, fair decimal odds and American odds are

$$d_{\mathrm{fair}} = \frac{1}{p}, \qquad A_{\mathrm{fair}} = \begin{cases} -100p/(1-p), & p > 1/2, \\ 100(1-p)/p, & p < 1/2. \end{cases} \tag{36}$$

At $p = 1/2$, even money is $+100$ (or $-100$). With win/push/loss probabilities summing to one, fair decimal *including refunded stakes on push* satisfies

$$d_{\mathrm{fair}} = 1 + \frac{p_{\mathrm{loss}}}{p_{\mathrm{win}}} = \frac{1 - p_{\mathrm{push}}}{p_{\mathrm{win}}}. \tag{37}$$

At offered decimal odds $d$, expected profit per dollar staked is

$$\mathrm{EV} = p_{\mathrm{win}}(d-1) - p_{\mathrm{loss}}, \tag{38}$$

with a push contributing zero. For two no-push quotes, implied raw probabilities $q_j = 1/d_j$ contain vigorish; a simple proportional no-vig approximation is $\widetilde{p}_j = q_j/(q_1+q_2)$. For 3-way prices include all three outcomes. More sophisticated de-vig choices should be versioned and tested.

## 8 Reconciling Elo and points ratings without contradictions

Elo provides an independent probability prior while the score model implies $p_S = \Pr(M > 0)$. Their discrepancy is useful diagnostic information: it can reveal a rating lag, incorrect scale, pace misspecification or roster data not yet reflected in scored games. The final published estimates must still be generated from *one* coherent score distribution.

A simple reconciliation uses a normal-equivalent Elo margin: if $\sigma_M$ is empirically fitted and $p_E$ is a calibrated Elo pregame win probability, define

$$\mu_M^{(E)} = \sigma_M \Phi^{-1}(p_E), \qquad \mu_M^* = (1-w)\mu_M^{(\mathrm{OD})} + w\mu_M^{(E)}, \quad 0 \le w \le 1. \tag{39}$$

Fit $w$ from held-out dates; do not pick it to maximize historical betting profit. To preserve the total forecast, reset $\mu_H^* = (\mu_T + \mu_M^*)/2$, $\mu_A^* = (\mu_T - \mu_M^*)/2$ *before* simulating scores. Ensure valid/nonnegative rates in the actual scoring generator. An alternative is a jointly trained latent-strength model with both Elo and offensive/defensive observations; this avoids blending separate probability estimates post hoc.

# 9 Illustrative end-to-end calculation (NFL)

The values below are *synthetic demonstration parameters*, not projections of a real matchup.

## 9.1 Pregame strength and score means

Suppose the home and away neutral Elo ratings are 1570 and 1510; home advantage is +45 Elo. Then

$$D = 1570 - 1510 + 45 = 105, \qquad p_E = \frac{1}{1+10^{-105/400}} \approx \boxed{64.6669\%}. \tag{40}$$

A separately fitted points-based model gives $r_H = +4.0$, $r_A = -1.0$ and $h = 1.6$ points, so $\mu_M = +6.6$. An offense/defense-plus-drive model predicts total $\mu_T = 47.4$ points. Equation (1) gives

$$\boxed{\mu_H = 27.0, \quad \mu_A = 20.4, \quad \mu_M = 6.6, \quad \mu_T = 47.4.} \tag{41}$$

Assume, for transparent *illustration only*, an approximate margin SD of $\sigma_M = 13.0$ and total SD $\sigma_T = 13.5$, fitted historically for a comparable set of forecasts. Then

$$\Pr(H \text{ wins}) \approx \Phi(6.6/13.0) = \boxed{69.4165\%}, \tag{42}$$

$$\Pr(T > 47.5) \approx \Phi((47.4 - 47.5)/13.5) = \boxed{49.7045\%}, \tag{43}$$

$$\text{Fair home moneyline} \approx \boxed{-227} \quad \text{(rounded American odds)}. \tag{44}$$

Elo alone yields 64.67% while this illustrative scoring model yields 69.42%. This is *not* a reason to report both as simultaneous official win probabilities: reconcile and calibrate the models as described in Section 8, then simulate one final scoring law. In this benchmark we retain the unblended scoring values to show the conversion math.

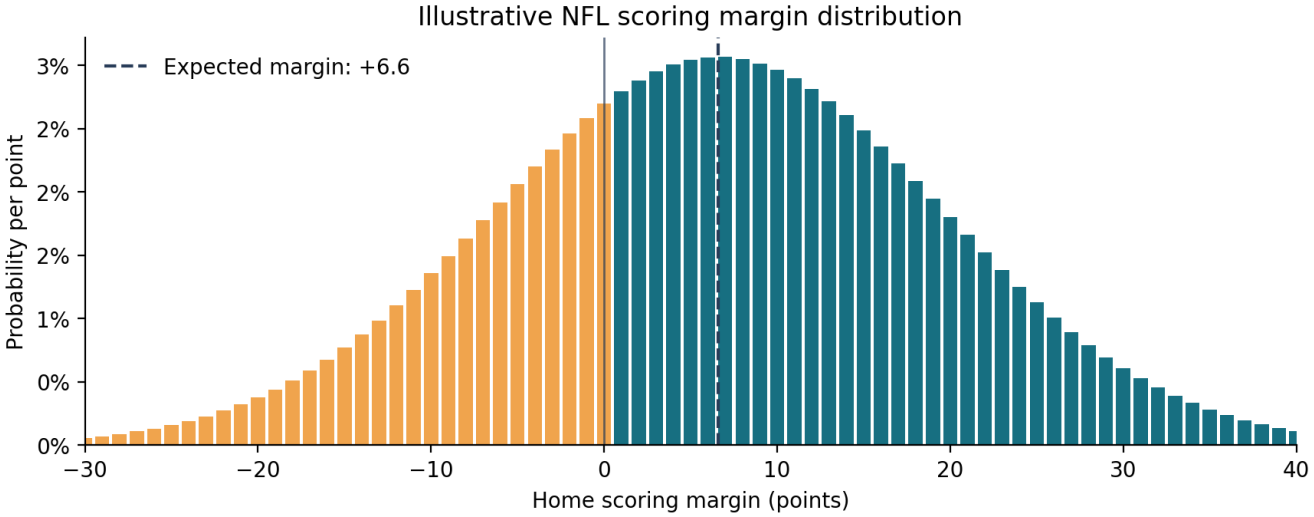



**Figure 2:** Discrete one-point bins constructed from an illustrative normal margin model; each bar is a probability, not a PDF density at a single point.

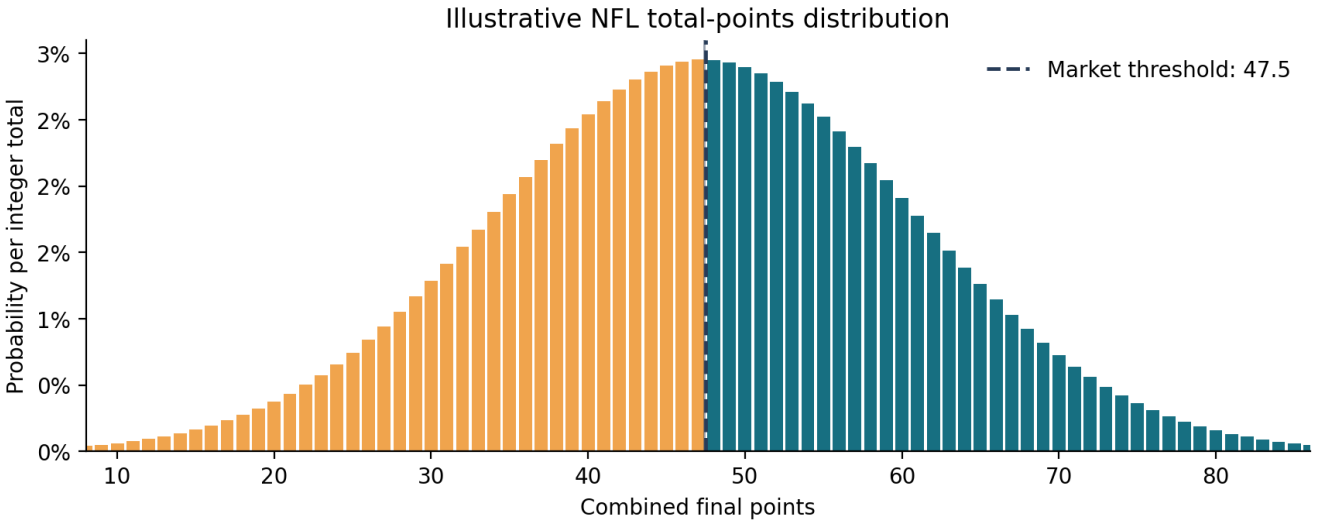



**Figure 3:** Approximate per-integer total probabilities. This distribution supports every candidate total line without refitting the matchup.

For any line $s$ or $L$, change only the event indicator applied to the *same* simulated score pairs. For instance a displayed home $-3.5$ spread pays if $M > 3.5$. A total of 47.0 has a genuine push state if final points equal 47; a total of 47.5 does not.

# 10 Parameter fitting and probabilistic calibration

## 10.1 Separate the predictive stages

1. Fit team-state model (Elo and/or dynamic points ratings) strictly sequentially.
2. Fit offense, defense, pace/drive parameters from opponent-adjusted observations; estimate effects of known roster and context.
3. Fit the *conditional distribution*, not just its mean: residual SD, tails, shared shocks, football scoring structure, OT.
4. Calibrate event probabilities on subsequent, unseen dates (e.g., logistic intercept/slope or isotonic). If calibrating moneyline and totals separately, reconcile through a coherent joint distribution or calibrate simulator parameters; avoid independent outputs that cannot arise from any one score law.

For a set of predicted game win probabilities $p_g$ and graded binary outcomes $w_g$, the binary log loss and Brier score are

$$\text{LogLoss} = -\frac{1}{G}\sum_g [w_g \log p_g + (1 - w_g)\log(1 - p_g)], \tag{45}$$

$$\text{Brier} = \frac{1}{G}\sum_g (p_g - w_g)^2. \tag{46}$$

For scores, compare error in $\mu_H, \mu_A, \mu_M, \mu_T$ and likelihood/CRPS of *actual* scores; for offered lines use calibration, binary proper scores and push-aware likelihood. Optimize over multiple objectives with controlled weights, not in-sample sportsbook ROI alone.

### 10.2 Estimating uncertainty robustly

If score residuals are heteroskedastic, let

$$\log \sigma_{M,g} = a_0 + a_1 \cdot \text{matchup imbalance} + a_2 \cdot \text{roster uncertainty} + a_3 \cdot \text{pace} + \cdots . \tag{47}$$

Use shrinkage and test whether the added complexity improves distributional scoring. For team-strength/forecast uncertainty draw $\theta^{(b)}$ from its posterior or use block bootstrap by week/season and a calibrated model ensemble. Report both *aleatoric* game randomness and *epistemic* parameter uncertainty; Monte Carlo sampling error is a third, usually much smaller, component.

### 10.3 Lagged covariates and updating clock

All training rows need fields `forecast_asof`, `source_published_at`, `scheduled_kickoff/tipoff`, `finalized_at`, and `model_version`. Injury announcements, odds snapshots and starting-lineup forecasts must be matched to the specific information cutoff. When a stat correction arrives, create a revised historical snapshot rather than silently altering the input to a previously recorded forecast.

## 11 Backtesting, benchmark competition and decision layer

### 11.1 Walk-forward evaluation

Use expanding/rolling windows in chronological order. Example: train on seasons up to year $y-1$, calibrate on a later non-overlapping block, and evaluate year $y$; repeat across multiple years. For weekly NFL/CFB and daily NBA runs, lock predictions at realistic pregame horizons (e.g., a day-ahead forecast versus a near-start forecast are separate products). No shuffling across future games.

| Component | Benchmark | Candidate enhancement |
|---|---|---|
| Moneyline | Elo alone with home factor | Elo + roster effects + joint score law |
| Spreads | Static home-edge + Massey/ridge ratings | Dynamic state-space O/D ratings |
| Totals | Rolling league/team scoring averages | Tempo + opponent-adjusted scoring + shared variance |
| Uncertainty | Constant normal residual SD | Conditional/discrete simulator + posterior draws |
| Calibration | Uncalibrated tail probability | Time-split calibration, reliability by league |
| Economic test | No action / naive lines | After-vig, timestamp-accurate edge monitoring |

### 11.2 Model diagnostics and failure modes

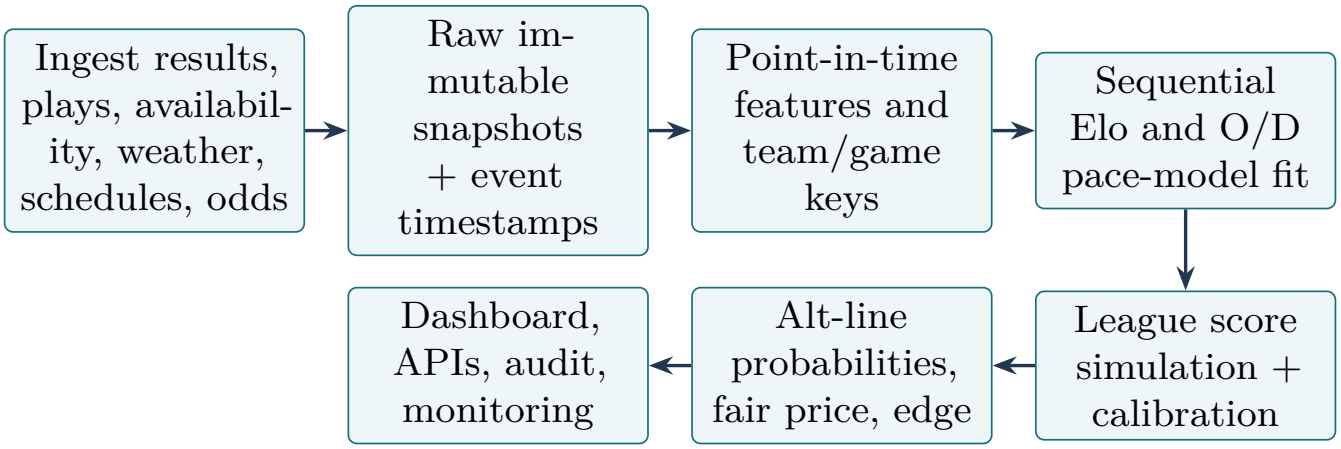
- **Reliability:** group $p$ into forecast-probability bins and compare predicted versus empirical frequency, including favored/underdog and season splits.
- **Marginal coverage:** check predicted 50%, 80% and 95% intervals for $M, T, Y_H, Y_A$; verify that realized push rates near integer lines match discrete predictions.
- **Time stability:** track drift by season, conference, rest bucket, roof/weather, injury severity and early-versus-late season.
- **Market comparison:** evaluate forecast MAE/log score against a time-aligned closing line as an external benchmark. Closing odds are not a guaranteed "truth" and cannot be used as a feature for an earlier forecast.
- **Execution realism:** store available quote, limits, latency, changes before acceptance, settlement conventions and actual filled odds. Do not infer arbitrage merely because model probability differs from a quote.

A conservative action test computes EV from Equation (38) using post-vig *offered payout* and model probabilities, then requires a material edge after uncertainty and execution costs. One possible robustness test is that a lower

model-confidence bound has positive EV after a predefined buffer; do not mistake a narrow Monte Carlo simulation interval for full model confidence. The report is methodological, not a guarantee of profitable bets.

# 12 Production system framework

## 12.1 Batch and pregame operations




## 12.2 Suggested modular Python project

```
sports_analytics/
  config/{nfl,nba,cfb}.yaml
  ingestion/{games,play_by_play,rosters,weather,odds}.py
  features/{asof_join,team_form,opponent_adjustment,pace}.py
  ratings/{elo,massey,offense_defense,state_space}.py
  models/{nfl_drives,nba_possessions,cfb_hierarchy}.py
  distribution/{simulate_scores,calibrate,market_events}.py
  validation/{walk_forward,calibration,score_diagnostics}.py
  pricing/{devig,fair_odds,edge}.py
  app/{api,dashboard}.py
  tests/
```

Use pandas and NumPy for transformations, SciPy for constrained least squares/distributions, scikit-learn or probabilistic-model libraries for calibration/hierarchical models, and Postgres for versioned forecasts, structured fixtures and odds. Persist raw game and play data as compressed, immutable Parquet. Docker and scheduled orchestration can drive nightly league-specific refreshes, with faster refreshes after materially important roster/news changes.

### 12.3 Suggested relational schema

| Table | Key fields (illustrative) |
|---|---|
| dim_team | team_id, league, season, conference, active_from/to |
| fact_game | game_id, scheduled_at, home/away_id, final_scores, result_status, ruleset_id |
| fact_odds | game_id, book, market_type, selection, line, decimal_odds, observed_at, received_at |
| fact_feature | game_id, forecast_asof, entity_id, feature_name, value, published_at |
| fact_rating | team_id, effective_asof, Elo, power, offense, defense, pace, uncertainty, model_version |
| fact_forecast | game_id, forecast_asof, version, mu_H, mu_A, p_H, diagnostics, input_hash |
| fact_marketprob | forecast_id, market, line, p_win, p_push, p_loss, fair_odds, quote_id |

Use stable team IDs across nickname changes, canonical home/away signs, reliable timezones and explicit ruleset IDs. Add unique constraints on snapshot keys and provenance hashes. Never overwrite a past forecast when the next odds or injury snapshot arrives.

### 12.4 Forecast procedure (minimum functional sequence)

```
for game in chronological_schedule:
    asof = chosen_forecast_cutoff(game)
    context = build_features(game, sources_published_before=asof)
    elo_prior = elo.predict(game, context) # NO postgame updates
    od_params = od.predict_team_scores(game, context)
    adjusted_params = reconcile(elo_prior, od_params, fitted_weights)
    scores = simulator[game.league].draw(adjusted_params, context,
                                         n=100_000, seed=fixed_seed(game))
    scores = settle_overtime_and_rules(scores, game.ruleset)
    forecast = derive_events_from_same_scores(scores, game.market_catalog)
    persist_immutable(game, asof, model_version, forecast, inputs_hash)
    # Only when result becomes official: update Elo + dynamic strength
```

The `100_000` draw count is an illustrative computational choice. Increase/reduce based on Monte Carlo standard error, latency and alt-line resolution; for very rare outcomes use variance-reduction or exact dynamic programming rather than relying on a modest brute-force sample.

### 12.5 Service endpoints and dashboard

`GET /games?league=...&date=...` lists fixtures; `GET /forecast/{game_id}?asof=...` returns model version, projected scores and expected margin/total; `GET /markets/{game_id}?type=...` supplies probability by threshold, fair prices, quote comparison and data timestamp.

A matchup page should have: two team cards (current Elo, power, O/D and matchup changes); expected final score; home/away moneyline probabilities; *separate bar charts of per-integer scoring margin and total-point probability*; hover/filters for alternate thresholds; model-vs-book quotes; and distribution uncertainty, roster flags and time of last data update. Every derived market should aggregate the same stored game-simulation identifier.

## 13 League-specific implementation roadmap

| Phase | NFL | NBA | College football |
|---|---|---|---|
| 1. Baseline | Elo + ridge margin + rolling total | Elo + margin + pace/O-D | Elo + priors + ridge margin |
| 2. Scores | Opponent-adjusted drive rates | Possession-based O/D | Hierarchical O/D per drive |
| 3. Distribution | Discrete drive and non-offense | Correlated score/pace and OT | Discrete scoring, unequal schedules, OT |
| 4. Context | QB, injuries, weather, travel | Minutes, rest, back-to-back, travel | Roster turnover, FBS/FCS, weather |
| 5. Validate | Weekly historical replay | Daily historical replay | Weekly replay; early-season priors |

**Recommended build order.** Deliver one league's full, correctly timestamped backtest before multiplying league-specific complexity. A reasonable first target is NFL or NBA: create reproducible Elo and offensive/defensive baselines, validate both margin and total distribution calibration, then integrate alternate market quote snapshots. Carry the same distribution/pricing API into CFB while retaining independent league parameters.

## 14 Critical caveats and launch checklist

- Elo differences do *not* uniquely determine totals; any model that maps Elo directly to O/U without a scoring/tempo baseline is underidentified.
- Fit $K$, Elo scale, home edge, offseason carry, ridge penalty, variance and blend weights independently per sport and era. Published historical third-party numbers are references, *not* current validated constants.
- Avoid correlated-label leakage: split on actual forecast time; preserve injury/lineup publication times; snapshot late corrections; do not train an early forecast using a later price.
- Point-margin models must address football score discreteness, OT and market grading. Team-score joint distributions are needed to avoid contradictory markets.
- A past backtest edge can be noise or execution bias. Evaluate scoring-rule accuracy, calibration, risk, limits and obtained prices before drawing any economic conclusions.

## 15 Sources and methodological anchors

The mathematics in this paper is a proposed model design, with illustrative assumptions and worked examples, not a claim that referenced publishers operate this exact combined system. The following sources support the cited definitions, historical Elo examples and data access points. URLs were checked on October 19, 2026.

## References

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